Synthetic Speech Overstates Code-Switching ASR: A Japanese–English Reality Check with Whisper and an On-Device Audio LM

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Abstract

Text-to-speech (TTS) synthesis is the standard workaround for the scarcity of labelled code-switching speech, but the cost of that substitution is rarely measured. We quantify it for Japanese–English. We fine-tune Whisper large-v3 with LoRA on a 350-clip synthetic corpus built from neural TTS and reach 2.48 % code-switching WER on its matched held-out split — a number that reads like a solved task. We then evaluate *that same merged checkpoint*, unchanged, on CS-FLEURS JA–EN read speech, where real human voices read code-switched sentences. It scores 36.3 % mixed error rate (MER) and still renders 27 % of English reference words outside Latin script. The adaptation does transfer — it improves on zero-shot Whisper large-v3 (40.9 % MER, 64.6 % script accuracy) — but the in-domain figure overstates deployable performance by a wide margin, and the relative error reduction collapses from 97.8 % in-domain to 11.2 % on real speech. Motivated by this gap, we adapt a second architecture: LFM2.5-Audio-1.5B-JP, a hybrid state-space/attention audio language model with a Conformer encoder. Applying LoRA to the *acoustic encoder* rather than the decoder alone lifts script accuracy from 37.4 % to 80.8 % and cuts MER to 27.0 % on the same frozen benchmark, surpassing both Whisper systems on code-switching while leaving Japanese monolingual accuracy intact (CER 6.6 % → 6.4 %). English monolingual recognition remains the outstanding weakness (37.1 % WER against Whisper's 4.2 %). We argue that script fidelity, not word error rate, is the metric that exposes synthetic-to-real degradation, and that encoder-side adaptation — not decoder prompting or decoder-only LoRA — is what actually buys code-switching capability.

Keywords: code-switching, speech recognition, low-resource adaptation, LoRA, synthetic training data, Japanese

1 Introduction

Code-switching — alternating between two languages within a single utterance — is routine in Japanese professional speech, where English technical vocabulary is embedded in Japanese syntactic frames. A speaker asks「次の meeting はいつ？」(“When is the next meeting?”) without pausing at the language boundary. Automatic speech recognition systems trained on monolingual corpora handle this poorly: they collapse output to one language, force English words into katakana, or drop the minority language entirely. The failure is not merely cosmetic. Downstream tasks — action-item extraction, search, summarisation — operate on the transcript, and a transcript that renders *deadline* as デッドライン has silently destroyed the string an enterprise pipeline was looking for.

The binding constraint on progress is data. Labelled, naturally-occurring code-switched speech is expensive to elicit and annotate, and for Japanese–English it is essentially unavailable at scale — a scarcity that has left the pair far behind Mandarin–English, where the SEAME corpus [12] has supported a decade of work. The field's standard response is to synthesise: draft code-switched sentences with a language model, render them with neural TTS, and fine-tune on the result. The approach is fast, cheap, perfectly aligned by construction, and increasingly common. What it costs in generalisation is almost never measured, because the same synthetic distribution supplies both the training and the test split.

This paper measures that cost. Our design is deliberately simple: train on synthetic speech, then evaluate the resulting checkpoint on real speech without touching it. We report both numbers for the same model, and the difference between them is the paper's primary result.

Measuring the gap requires the right instrument. Word error rate is a poor one here, because the dominant code-switching failure — rendering an English word in katakana instead of Latin script — is scored as a single substitution, indistinguishable from any ordinary misrecognition. We therefore adopt *script accuracy*: the fraction of English reference words that the system renders in Latin script rather than katakana, another script, or not at all. It isolates the code-switching behaviour from general recognition quality, and it turns out to be the metric on which synthetic training and real speech diverge most sharply.

Our contributions are:

1. **A quantified synthetic-to-real gap for code-switching ASR.** A Whisper large-v3 LoRA checkpoint that achieves 2.48 % CS-WER on its matched synthetic test split retains only 11.2 % relative MER improvement over zero-shot on real code-switched read speech, against 97.8 % in-domain (Section 8).
2. **Evidence that encoder adaptation, not decoder adaptation, drives code-switching gains.** On LFM2.5-Audio, decoder-side LoRA with data augmentation moves script accuracy by 6.2 points; adding LoRA to the Conformer encoder adds a further 18.2 points (Section 7).
3. **A frozen, reusable JA–EN evaluation protocol** combining a code-switched test set with matched monolingual controls drawn from the same sentences and speakers, so that forgetting is separable from domain shift (Section 4).
4. **A deployable on-device result**: 80.8 % script accuracy from a 1.5 B-parameter model with 8.16 M trainable parameters, running faster than real time on an 8 GB laptop GPU (Section 9).

We also report a negative result plainly: our best code-switching model remains far behind Whisper on English monolingual audio (37.1 % versus 4.2 % WER). We consider this the most important open problem the work leaves behind, and we do not discount it.

2 Background and Related Work

2.1 Multilingual and code-switching ASR

End-to-end architectures displaced HMM-GMM pipelines [19] through CTC [4], attention-based encoder-decoders [2], and the Conformer [5], whose interleaving of self-attention with convolution remains the dominant audio encoder design. Whisper [20] trained an encoder-decoder Transformer on 680 k hours of weakly supervised audio across 96 languages, producing a model whose multilingual prior is strong enough that adaptation to a new task often requires hundreds rather than thousands of hours. Massively multilingual efforts [18, 27] extended the same scaling argument further.

Code-switching has been studied linguistically for decades [17, 13]. The Matrix Language Frame model [13] holds that one language supplies the morphosyntactic frame while the other contributes content morphemes; in Japanese–English discourse Japanese is consistently the matrix language, with English nouns and technical terms inserted at positions Japanese syntax permits. This asymmetry matters for ASR: English lexical items occur in acoustic contexts shaped by Japanese prosody and co-articulation, distributions a monolingual English model has never observed.

Computational work has concentrated on Mandarin–English, where SEAME [12] provides real conversational data. Shan et al. [21] added language-aware attention; Winata et al. [24] applied meta-transfer learning for few-shot adaptation. Both required either large labelled code-switching corpora or expensive meta-training. Peng et al. [16] showed that Whisper's pre-training is strong enough that prompting alone recovers substantial code-switching ability, reducing the in-domain data requirement by orders of magnitude. Japanese–English has received comparatively little attention, and CS-FLEURS [25] is the first broad-coverage benchmark to include the pair.

2.2 Parameter-efficient fine-tuning

LoRA [7] reparameterises a weight update as a low-rank product $\Delta W = BA$ with $B \in \mathbb{R}^{m\times r}$, $A \in \mathbb{R}^{r\times n}$ and $r \ll \min(m, n)$, freezing W and training only A and B. Because B is zero-initialised, $\Delta W = 0$ at the start of training and the adapted model begins from the full strength of the pre-trained weights — a property that matters when the adaptation set is small enough that a random perturbation would be destructive. Alternatives include adapter modules [6], prefix tuning [10], and prompt tuning [9]; LoRA is preferred here because the adapter can be merged into the base weights after training, incurring no inference-time overhead. Aghajanyan et al. [1] give the theoretical motivation: fine-tuning updates occupy a subspace whose effective dimension is far below the ambient one, and scales with task complexity and dataset size.

2.3 Synthetic speech as training data

Using TTS to manufacture training audio is well established for low-resource ASR, and augmentation with recorded noise [22], spectrogram masking [15], and speed perturbation [8] are standard complements. For code-switching specifically, synthesis is unusually attractive: a language model can generate switch points on demand, and the resulting transcript is exact by construction, eliminating the inter-annotator disagreement that plagues manual code-switching annotation [26]. CS-FLEURS [25] itself uses synthesis for the majority of its test sets, reserving real voices for a 14-language subset.

The literature is correspondingly thin on what synthesis costs. Yılmaz et al. [26] compared augmentation strategies for code-switching but did not isolate synthetic-versus-real training audio. To our knowledge, no prior work evaluates a single code-switching checkpoint on both a matched synthetic test set and real speech, which is precisely the comparison we report.

3 Corpora

3.1 The synthetic training corpus

The training corpus contains 350 clips: 100 code-switching, 200 Japanese-only, and 50 English-only. Source sentences were drafted with a large language model and rendered with Microsoft Azure neural TTS — `en-US-JennyNeural` for English units and a `ja-JP` female neural voice for Japanese — at 24 kHz mono. Code-switching clips concatenate per-unit renderings in language-alternation order with approximately 1.2 s of silence at each junction; monolingual clips use 300 ms. Because each clip is rendered *from* its transcript, audio and reference are exactly aligned and annotation noise is zero.

A 70/30 utterance-level split with a fixed seed gives 245 training and 105 test clips (Table 1). Training audio is augmented with MUSAN [22] noise at 20 dB and 15 dB SNR, giving 2×245 augmented variants plus the originals for 735 training samples.

Two properties of this corpus are essential to the argument. First, it is *convenient*: it was assembled in days, at negligible cost, with perfect labels. Second, it is *narrow*: one voice per

Table 1: Synthetic corpus composition. The 70/30 split is utterance-level with a fixed seed; augmentation applies to the training partition only.

Category	Train	Test	Total	Augmented train
Code-switching	70	30	100	210
Japanese-only	140	60	200	420
English-only	35	15	50	105
Total	245	105	350	735

language, no disfluencies, no prosodic switch cues, no channel variability, and inter-unit silences that are a synthesis artefact rather than a property of bilingual speech. These are exactly the properties a practitioner accepts when choosing synthesis, and exactly the ones whose cost we set out to measure.

3.2 The real-speech evaluation benchmark

Evaluation uses CS-FLEURS [25], built over FLEURS [3]. We take the entire Japanese–English `read/test` slice: 196 utterances in which *human speakers read code-switched sentences*. The sentences are constructed by aligning FLEURS translations and swapping spans, so the switch points are synthetic; the acoustics are not. This is the axis on which our training corpus is weakest, and therefore the axis worth testing.

Critically, no CS-FLEURS data of any split is used in training. The training corpus is generated independently, so the full 196-utterance slice is a clean held-out set and the same fixed utterances score every system in this paper.

3.3 Monolingual controls

We add two 200-utterance monolingual sets from FLEURS `ja_jp` and `en_us` test. Because CS-FLEURS is constructed by span-swapping over the *same* FLEURS sentences, in the same style and register, these controls are matched rather than merely adjacent: a monolingual regression after fine-tuning is attributable to the adaptation, not to a domain shift between benchmarks. This lets us separate "learned to code-switch" from "forgot how to do Japanese".

4 Evaluation Methodology

4.1 Metrics

Mixed-script transcription defeats naive word-level scoring, because Japanese has no orthographic word boundaries. We therefore score four quantities, each over an explicitly defined token stream, with identical normalisation (NFKC, lowercasing, punctuation mapped to space, English fillers removed) applied to references and to every hypothesis.

Script accuracy (ScriptAcc) is the headline. Of the English words in the reference, it is the fraction the system renders in Latin script:

$$\text{ScriptAcc} = \frac{|\{\, w \in R_{\text{en}} : \hat{w} \text{ is Latin-script} \,\}|}{|R_{\text{en}}|} \tag{1}$$

where R_{en} is the set of English reference words. A word forced into katakana, rendered in another script, or dropped all count against it. ScriptAcc answers the question a bilingual user actually asks — *did it keep my English as English?* — and, unlike WER, it is unaffected by ordinary recognition errors elsewhere in the utterance.

Mixed error rate (MER) is an edit rate over a mixed token stream, taking one token per Japanese character and one per English word, which avoids dependence on a Japanese word segmenter. **JA-CER** is character error rate over the Japanese-only stream and **JA-WER** its word-level counterpart via `fugashi` segmentation; **EN-WER** is word error rate over the Latin stream. The monolingual metrics are forgetting controls, not headline results.

4.2 A note on metric commensurability

The dissertation-stage experiments on the synthetic corpus report **CS-WER**: whitespace-delimited word error rate over the mixed reference. The real-speech benchmark reports MER. *These are different metrics, computed over different token streams, on different test sets, and their absolute values must not be subtracted or divided across the two settings.* We are explicit about this because the temptation to write "2.48 % became 36.3 %, a 15× degradation" is strong and the statement would be unsound.

What *is* sound is a comparison of each system against a common baseline *within* each setting. On the synthetic set, LoRA reduces CS-WER from 111.57 % (zero-shot) to 2.48 %, a 97.8 % relative reduction. On real speech, the same checkpoint reduces MER from 40.9 % (zero-shot) to 36.3 %, a 11.2 % relative reduction. Both are relative improvements over the same base model, each measured with one metric on one set, and it is their *ratio of relative gains* — not the raw error rates — that quantifies how much of the adaptation survives contact with real speech.

4.3 Inference protocol

Every system is evaluated under one protocol. LFM systems use the system prompt `"Perform ASR."`, matching their training data; the base model is strongly prompt-sensitive, so this is held fixed across conditions. Whisper systems run through `transformers` on GPU with automatic language detection, `repetition_penalty` = 1.3 and `no_repeat_ngram_size` = 3 to suppress the decoding loops both models exhibit on code-switched input. Metrics are pooled corpus-wide rather than averaged per utterance, avoiding the instability short utterances introduce.

5 Whisper LoRA Adaptation

We adapt Whisper large-v3 (1.5 B parameters, 32-layer encoder and decoder, $d_{\text{model}} = 1280$, 128-mel front end) with LoRA on the query and value projections of the decoder attention layers, freezing the encoder entirely. The configuration is $r = 8$, $\alpha = 16$, dropout 0.1, learning rate 1×10^{-4} held constant, effective batch size 16 (4 per device × 4 accumulation steps), bf16 precision. This yields roughly 3.9 M trainable parameters, 0.26 % of the base model.

Freezing the encoder encodes a hypothesis: that Whisper's multilingual pre-training already supplies adequate acoustic representations for both languages, and that code-switching adaptation is principally a matter of adjusting the decoder's language prior. Section 6 tests that hypothesis directly, and finds it wanting.

Two implementation details proved decisive. Japanese language forcing via `forced_decoder_ids` must be applied at inference; without it the untrained decoder emits predominantly English for code-switched audio, since English dominates Whisper's pre-training distribution. And label padding positions must be masked to −100 so that cross-entropy ignores them — with the leading BOS token stripped, since the trainer prepends it during teacher forcing.

Development proceeded in four phases. Phase 1 established Small and Medium baselines. Phase 2 was invalidated by a data-loading defect that read an empty column, yielding empty references and meaningless error rates; we report it here because the failure is instructive and because its checkpoints must not be mistaken for results. Phase 3 fixed the defect and compared three large-v3 configurations (Table 2). Phase 4 extended the winner to 15 epochs, warm-started from the Phase 3 checkpoint.

The final checkpoint is merged into the base weights and published as a standalone model, which is the artefact evaluated on real speech in Section 8. No further training of any kind is

Table 2: Phase 3 configuration comparison on the synthetic test split. Learning rate dominates rank: both 5×10^{-5} runs fail to converge within the budget.

Configuration	LR	r	Modules	Best ep.	CS-WER (%)
`v3_r8_1e4`	1×10^{-4}	8	q, v	8	**6.61**
`v3_r4_5e5`	5×10^{-5}	4	q, v	7	85.95
`v3_r8_attn4`	5×10^{-5}	8	q, v, k, out	8	45.45

applied to it.

6 LFM2.5-Audio Adaptation

The synthetic-to-real result motivated a second system built on a different inductive bias. LFM2.5-Audio-1.5B-JP [11] pairs a hybrid state-space/attention language backbone with a 17-layer Conformer audio encoder and an MLP adapter, at 1.5 B parameters total. Unlike Whisper it is designed for on-device deployment, and unlike Whisper its encoder is small enough to adapt cheaply.

6.1 Where to put the adapters

The package ships no LoRA support, so adapters are attached manually via PEFT. This freedom is the point of the experiment. We compare two placements:

Decoder-side only. LoRA on the language backbone — attention projections `q_proj`, `k_proj`, `v_proj`, `out_proj`, the state-space `in_proj`, and feed-forward `w1`, `w2`, `w3` — with $r = 8$, $\alpha = 16$, dropout 0.05. This is the direct analogue of the Whisper recipe and yields approximately 4.7 M trainable parameters (0.32 %).

Encoder *and* decoder. The same backbone targets plus LoRA on the Conformer's attention (`linear_q`, `linear_k`, `linear_v`, `linear_out`) and feed-forward (`linear1`, `linear2`) projections. The pointwise convolutions are `Conv1d` and are left alone. This gives 8 155 136 trainable parameters (0.5579 %).

Encoder and backbone adapters are optimised as two AdamW parameter groups with separate learning rates: 3×10^{-5} for the backbone and 6×10^{-6} — five times lower — for the encoder. The asymmetry is deliberate. The encoder carries phonological representations learned from far more audio than we can supply, and an aggressive update destroys them; the backbone's language prior is what we actually intend to move.

6.2 Training data

Training uses the synthetic corpus of Section 3, extended with MUSAN augmentation at three levels (noise at 20 dB and 13 dB, music at 20 dB) and a task taxonomy that pairs each audio type with a directed system prompt (`"Perform ASR."`, `"Perform ASR in Japanese."`, `"Perform ASR in English."`). The best-performing run additionally mixes in FLEURS-balanced monolingual data to counter English forgetting.

[TODO: The best configuration reported in Table 5 (rank-32 encoder LoRA with the FLEURS-balanced mix) is not covered by the project training log, which stops at the preceding run. Recover and document its step count, learning rates, data proportions, and validation curve before submission — the paper's strongest number currently has no reproducible recipe.]

Table 3: Whisper LoRA results on the *synthetic* held-out split ($n = 105$). RTF and VRAM are single-sample inference measurements. These numbers are in-domain: training and test share one TTS distribution and one voice per language.

		WER (%) ↓			CER (%) ↓			
Model	Params	CS	JA	EN	CS	JA	EN	VRAM
Whisper Small	244 M	9.92	10.00	1.90	0.99	1.09	1.01	0.54 GB
Whisper Medium	769 M	6.61	**0.00**	**0.32**	0.40	**0.00**	**0.05**	1.59 GB
Whisper large-v3	1.5 B	**2.48**	0.83	2.22	**0.40**	0.05	1.61	3.13 GB
large-v3, zero-shot		111.57	100.00	1.59	–	–	–	–

Table 4: Phase 4 training trajectory, `v3_r8_1e4` extended to 15 epochs. Code-switching competence emerges abruptly between epochs 5 and 7, well after Japanese monolingual accuracy has converged.

Epoch	0	1	4	7	8	10	12	15
CS-WER (%)	111.57	104.96	71.90	7.44	5.79	4.96	4.96	**2.48**
JA-WER (%)	100.00	100.00	4.17	3.33	2.50	0.83	0.83	0.83
EN-WER (%)	1.59	1.90	1.90	1.90	1.90	1.59	2.22	2.22

7 Results

7.1 In-domain performance on synthetic speech

Table 3 reports all three Whisper scales on the matched synthetic test split. Error rates are low across the board and fall monotonically with model size on code-switching audio, reaching 2.48 % CS-WER for large-v3. Read in isolation, this table says the task is solved.

The training trajectory (Table 4) shows a phase structure worth noting: Japanese WER collapses from 100 % to 4.2 % by epoch 4 while CS-WER remains above 70 %, and only between epochs 5 and 7 does CS-WER fall steeply. The model learns Japanese phonology first and language switching second. Terminating on a fixed patience at epoch 5 would have recorded this configuration as a failure.

7.2 Performance on real code-switched speech

Table 5 is the paper's central table. Every system is scored on the same frozen 196-utterance CS-FLEURS slice and the same monolingual controls.

Four findings follow.

The base models disagree sharply on code-switching. Whisper large-v3 zero-shot keeps 64.6 % of English words in Latin script; LFM2.5-Audio-1.5B-JP keeps 37.4 %. The Japanese-specialised model is markedly worse at preserving English, which is consistent with its training emphasis and is the deficit our adaptation targets.

Encoder LoRA is what moves script accuracy. Decoder-side LoRA with data augmentation raises ScriptAcc from 37.4 % to 43.6 % (+6.2). Adding LoRA to the Conformer encoder — holding the data fixed — raises it to 61.8 % (+18.2), roughly three times the gain from decoder adaptation. This is a clean single-variable comparison and it directly contradicts the frozen-encoder assumption underlying the Whisper recipe in Section 5. *We note that the final row changes two things at once* — rank 8 → 32 and the addition of the FLEURS-balanced mix — so its +19.0 further gain cannot be attributed to either factor alone.

Code-switching gains need not cost Japanese. The best configuration ends at JA-CER 6.4 % against the base model's 6.6 % and JA-WER 7.2 % against 7.6 % — marginally better, not worse. Because the monolingual controls are drawn from the same sentences and speakers as the code-switching set, this is evidence of genuinely retained capability rather than an artefact

Table 5: Results on the frozen real-speech benchmark: CS-FLEURS JA–EN `read/test` (n = 196, code-switching) with matched FLEURS monolingual controls (n = 200 each). ScriptAcc is the headline code-switching metric; the monolingual columns are forgetting controls. Whisper large-v3 + LoRA is the *same merged checkpoint* that scores 2.48 % CS-WER in Table 3.

	Code-switching		Monolingual controls		
System	ScriptAcc ↑	MER ↓	JA-CER ↓	JA-WER ↓	EN-WER ↓
LFM2.5-Audio-1.5B-JP (zero-shot)	37.4	67.3	6.6	7.6	68.5
Whisper large-v3 (zero-shot)	64.6	40.9	**6.0**	7.9	**4.2**
This work					
Whisper large-v3 + LoRA (synthetic-trained)	73.0	36.3	–	–	–
LFM + LoRA, augmentation only	43.6	54.3	8.2	8.7	80.2
LFM + LoRA, + Conformer encoder LoRA	61.8	54.1	9.4	10.2	73.1
LFM + LoRA, r=32 encoder + FLEURS mix	**80.8**	**27.0**	6.4	**7.2**	37.1

– = not measured. See Section 10.

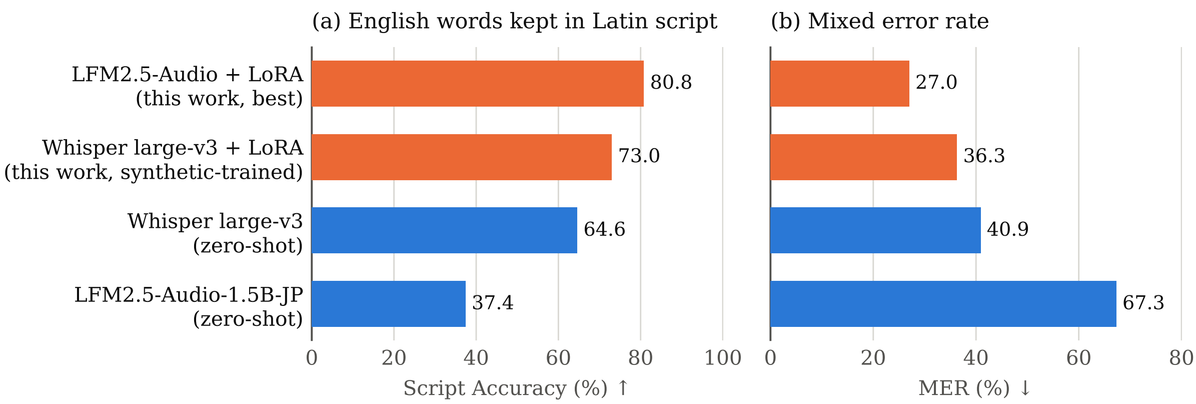

Figure 1: Systems on the frozen real-speech benchmark (CS-FLEURS JA–EN, n = 196). ScriptAcc and MER differ in scale and direction and are therefore plotted on separate axes, never shared. Orange marks systems from this work.

of benchmark mismatch. The intermediate rows show the mechanism is not automatic: the two weaker configurations both degrade Japanese (to 8.2 % and 9.4 % CER) before the balanced data mix restores it.

English monolingual recognition is the unresolved failure. Every LFM configuration is far behind Whisper on English-only audio. The base model starts at 68.5 % EN-WER; decoder-only LoRA makes it *worse* (80.2 %); the best configuration recovers to 37.1 %, roughly half the base error but nearly nine times Whisper's 4.2 %. A system that keeps English in Latin script but transcribes English audio at 37 % WER is useful for bilingual meetings and unsuitable as a general English recogniser. We state this plainly rather than confining it to a limitations paragraph.

7.3 Ablations on the synthetic corpus

The ablations below were run on Whisper Small for tractability and are reported for what they are: evidence about *the synthetic training regime*, not about real-speech performance. They are informative for practitioners building similar corpora and should not be read as generalising to natural speech.

Three observations. *Noise augmentation is not optional*: without it CS-WER sits at the untrained baseline, and the second SNR level produces a tenfold improvement — a threshold effect, not a gradual one. *Rank is non-monotonic*: r = 16 is optimal for Small, with r = 32

Table 6: Ablations on the synthetic corpus (Whisper Small). Each varies one factor with all others held fixed. Note the scale: without augmentation the model does not learn the task at all.

Condition	CS-WER (%)	CS-CER (%)	JA-WER (%)	EN-WER (%)
(a) Noise augmentation level				
None	271.90	55.71	33.33	2.54
SNR 20 dB	171.90	23.17	21.67	2.86
SNR 20+15	17.36	2.38	15.83	2.22
SNR 20+15+10+5	**13.22**	**1.06**	**11.67**	**1.90**
(b) LoRA rank (q, v projections)				
$r = 4$	55.37	4.88	24.17	2.86
$r = 8$	16.53	2.64	21.67	2.22
$r = 16$	**14.05**	**2.24**	17.50	2.22
$r = 32$	18.18	2.24	**14.17**	**1.90**
(c) Training data composition				
CS only	255.37	56.77	95.83	4.13
JA only	682.64	172.34	27.50	2.86
CS + JA	23.97	**2.97**	**15.83**	**1.90**
CS + JA + EN	**17.36**	2.51	17.50	2.86

degrading through overfitting on 70 code-switching training clips, consistent with the intrinsic-dimensionality account of Aghajanyan et al. [1]. *Monolingual data is necessary*: training on code-switching audio alone leaves CS-WER at the untrained baseline, and adding English-only clips improves code-switching performance despite contributing only 35 training utterances. For corpus builders, the practical implication is that monolingual recordings — cheap, and available from existing corpora [23, 14] — carry disproportionate value relative to expensive code-switched elicitation.

8 The Synthetic-to-Real Gap

We now compare the two settings for a single checkpoint. The model is Whisper large-v3 with the r=8, 1×10^{-4}, 15-epoch adapter merged into the base weights. It is evaluated on its matched synthetic test split and on CS-FLEURS JA–EN with no intervening training, no recalibration, and no prompt changes beyond the shared protocol of Section 4.

Following Section 4.2, we compare relative improvements over the same zero-shot base model within each setting:

Setting	Zero-shot	+ LoRA	Relative gain
Synthetic TTS test split ($n = 105$), CS-WER	111.57	2.48	97.8 %
CS-FLEURS JA–EN ($n = 196$), MER	40.90	36.30	11.2 %

The adaptation transfers, but weakly. Nearly all of the in-domain improvement is attributable to fitting the synthetic distribution rather than to acquiring code-switching capability that survives a change of speaker and channel. On ScriptAcc the picture is the same: the fine-tuned checkpoint reaches 73.0 % against zero-shot Whisper's 64.6 %, an 8.4-point gain, and it still fails to preserve Latin script for more than a quarter of English reference words.

Why the degradation is this large follows from Section 3. The training corpus has one voice per language, so the model may minimise loss by keying on voice-specific spectral regularities rather than on language-general acoustics. It has fixed inter-unit silences at switch points, so the model can learn to treat silence as a switch cue — a cue that does not exist in the fluent

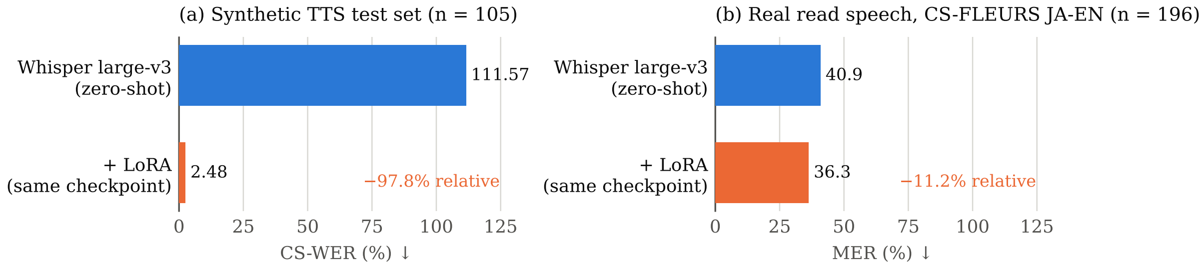

Figure 2: The same merged checkpoint on two test sets. Each panel has its own metric and axis — CS-WER and MER are not commensurable (Section 4.2) — so the comparison is between the *relative* gains over a shared zero-shot baseline, not between raw values.

Table 7: Inference on an RTX 4060 Laptop GPU (8 GB), averaged over six clips (mean duration 11.1 s). RTF is inference time divided by audio duration; all three models run far below real time and fit concurrently in 8 GB.

Model	Params	Precision	Model VRAM	Mean inference	Mean RTF
Whisper Small	244 M	fp16	478 MB	0.192 s	0.020
Whisper Medium	769 M	fp16	1458 MB	0.445 s	0.048
Whisper large-v3	1.5 B	bf16	2945 MB	0.651 s	0.064

switches of CS-FLEURS. It has no disfluencies, no prosodic switch marking, and no channel variation. None of these are exotic failure modes; they are the direct, predictable consequences of the corpus design, and the size of the gap is our estimate of what they cost.

We draw two conclusions for practice. First, **a synthetic held-out split is not a validation set for a synthetically-trained code-switching model**; it measures fit to the synthesiser. Any claim about deployable performance needs real speech, even a small quantity of it. Second, **script accuracy exposes the gap better than word error rate**. It moves 8.4 points where relative MER gain collapses from 97.8 % to 11.2 %, and it names the specific failure — katakana substitution — that makes a transcript unusable downstream while barely registering as a WER penalty.

9 Deployment

The motivating application is on-device transcription of privacy-sensitive bilingual meetings, where sending audio to a cloud API is prohibited. We therefore measure inference on consumer hardware: an NVIDIA RTX 4060 Laptop GPU (8 GB VRAM, Ada Lovelace) in a Dell G15, with merged checkpoints loaded from local disk, one warm-up pass, fp16 for Small and Medium and bf16 for large-v3.

All three models run one to two orders of magnitude faster than real time, and all three load simultaneously in roughly 4.9 GB — meaning a side-by-side comparison interface fits on a mid-range laptop. Latency, not throughput, would bind in a live-captioning deployment.

Two generation artefacts appeared during benchmarking and are worth recording, since aggregate error rates conceal them. Whisper large-v3 entered a repetition loop on one code-switching clip (`何か adviceadad…`), and Whisper Medium hallucinated a Japanese transcript for English-only audio. Both are known autoregressive failure modes, both are mitigated by the repetition penalty and n-gram blocking in our evaluation protocol, and both indicate that a deployed system needs decoding safeguards independent of its error rate on a benchmark.

10 Limitations

The real-speech benchmark is small and single-speaker. CS-FLEURS JA–EN `read/test` contains 196 utterances from one speaker. Results are therefore speaker-specific, and we report no confidence intervals; bootstrap resampling over utterances is the obvious remedy and has not been run. Additionally, CS-FLEURS switch points are generated by span-swapping, making them denser and more regular than natural conversational switching, so absolute error rates on it run high relative to what naturalistic speech would produce.

One incomplete row. The Whisper large-v3 + LoRA system has not been scored on the FLEURS monolingual controls, so its forgetting profile is unknown and Table 5 carries three empty cells. The evaluation harness supports this directly; the run has simply not been performed.

One undocumented recipe. As flagged in Section 6, the training configuration for the best-performing LFM row is not recorded in the project logs. Until recovered, that row is a measurement without a method.

Confounded final ablation step. The transition from 61.8 % to 80.8 % ScriptAcc changes LoRA rank and training data simultaneously. The individual contributions are not separable from the runs performed.

No seed variance. Every configuration was trained once. Reported differences below roughly two points should not be treated as reliable orderings.

Synthetic-corpus ablations do not transfer automatically. The augmentation, rank, and composition results in Table 6 were measured in the synthetic regime, and the central finding of this paper is precisely that conclusions drawn there need not hold on real speech.

11 Conclusion

We set out to measure what synthetic training data costs in code-switching ASR, and found the cost to be large. A Whisper large-v3 LoRA checkpoint that reduces code-switching error by 97.8 % relative on its matched synthetic test split retains only 11.2 % relative improvement when the same checkpoint meets real code-switched read speech. The adaptation is genuine but modest; the in-domain number is mostly a measurement of fit to the synthesiser. Practitioners should treat a synthetic held-out split as a training diagnostic rather than as evidence of deployability, and should adopt script accuracy alongside word error rate, because it names the failure — English forced into katakana — that ruins a transcript downstream while barely moving WER.

The constructive half of the paper is that architecture placement matters more than we expected. Adapting the Conformer encoder of LFM2.5-Audio, rather than the language backbone alone, produced roughly three times the script-accuracy gain of decoder-side adaptation, and the best configuration reached 80.8 % ScriptAcc and 27.0 % MER — surpassing Whisper large-v3 on code-switching — while leaving Japanese monolingual accuracy intact and running faster than real time on an 8 GB laptop GPU. The frozen-encoder convention inherited from text-domain LoRA practice appears to be the wrong default when the task is acoustic language alternation rather than a shift in output style.

What remains open is English. Our best code-switching system transcribes English-only audio at 37.1 % WER against Whisper's 4.2 %, and closing that gap without surrendering script accuracy is the natural next problem. Beyond it, the priorities that follow from this work are a speaker-diverse real-speech evaluation set with confidence intervals, the separation of rank from data mixture in the final ablation, and a direct test of whether encoder-side adaptation confers the same advantage on Whisper that it does on LFM2.5-Audio.

Reproducibility

Training, preprocessing, and evaluation code, the scoring harness with unit tests, and the CS-FLEURS and FLEURS loaders are available in the project repository. The merged Whisper checkpoint evaluated in Section 8 and the LFM adapters are published on the Hugging Face Hub. Raw prediction files from the real-speech evaluation are not currently version-controlled; the numbers in Table 5 are transcribed from project evaluation logs, and regenerating them from the published checkpoints is a documented single-command operation.

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